

Feature 233energy + causal steering

Basic/acidic after an LSE/LSEE motif

Top activating sites

LLSEEE[R]KEREKI	12.3
ELSEEE[K]EEMEKA	12.2
EEEKKK[K]EEEEKR	12.2
KLSEEE[K]KKAEEE	12.1
TLSEEE[K]EKLKKE	12.1

Fires in	46,327 / 64,998 designs
Energy	−13.5 kcal/mol (acti., q=0.011)
Steering	+2.8pp vs random (p<0.001, n=2184)

The only feature real on every arm tested.

Feature 6073energy only

Leu preceded by acidic (D/E) — 6 of top 12

Top activating sites

KRAVQQ[L]RACGVL	73.5
QAFDEE[L]AFYGLV	36.1
SAYDEE[L]AFFGLI	35.5
PLYEEE[L]AFWGID	34.4
LDKLQV[E]RERGIT	21.7

Fires in	273 / 64,998 designs
Energy	−30.1 kcal/mol (pres., q=0.011)
Steering	+1.2pp vs random (p=1.00, n=84)

Largest energy effect of any feature; steering null.

Feature 11326energy only

Basic (R/K) + Pro, before acidic/Pro-rich

Top activating sites

KGDPGS[P]GPPGPA	18.4
EALRKA[R]ADPSVY	15.5
SADGKE[K]SPPAYL	15.1
TTAPDI[R]EDPKAF	15.1
RGEQGT[P]GPPGPA	14.2

Fires in	64,997 / 64,998 designs
Energy	−12.9 kcal/mol (acti., q=0.005)
Steering	+0.8pp vs random (p=0.21, n=1680)

Fires in 64,997/64,998 designs — steer it up, you cannot introduce it.